

TRM Project Status Report

Less is More: Recursive Reasoning with Tiny Networks

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1 Purpose of This Document

This document is a development checkpoint for the TRM project.

Its purpose is to preserve important decisions, experiments, validation results, and current project status so that development can continue in a new chat, with another AI assistant, or with another team member without reconstructing previous context.

2 Project Overview

The project is based on Tiny Recursive Model (TRM).

The final research direction is:

Selective Attention over the Latent History of TRM

The overall strategy is:

- Establish a reliable baseline implementation of original TRM.
- Implement HistoryAttention as a separate modification.
- Compare the modified model against the baseline.
- Perform ablation studies and analysis.

3 Phase 0: CPU Foundation

3.1 Objective

The goal of Phase 0 was to build a reproducible and stable foundation before any architectural modification.

The objectives were:

- Enable CPU-only execution.
- Validate the training pipeline.
- Validate dataset loading.
- Verify model execution.
- Create reproducible checkpoints.
- Establish Git workflow.

3.2 Validated Environment

The verified environment:

- Conda environment: `trm-cpu`
- Python: 3.12.13
- PyTorch: 2.7.0+cpu
- CUDA availability: False
- CPU threads: 8

3.3 Repository Structure

Current structure:

```
TRM Project

config
dataset
evaluators
models
scripts
utils
data
checkpoints
experiments
```

3.4 Smoke Test Dataset

A minimal dataset was created:

```
data/sudoku-tiny
```

Purpose:

- Pipeline verification only.
- Not used for scientific experiments.

Dataset size:

- Training examples: 8
- Test examples: 4

3.5 Smoke Run Validation

The following components were successfully validated:

- Model loading.
- Dataset loading.
- Training loop.
- Evaluation loop.
- Checkpoint creation.

Checkpoint:

```
checkpoints/  
  
trm-cpu-smoke/  
  local-smoke/  
    step_16
```

4 Git Workflow

Current branches:

```
main  
  
codex/cpu-foundation  
  
phase1-baseline
```

Development decision:

- Phase 0 foundation was isolated from experimental development.
- Phase 1 development started on a separate branch.

5 Phase 1: Baseline Stabilization

5.1 Objective

The goal of Phase 1 is to create a scientifically meaningful baseline.

The key question:

What performance does the original TRM achieve under our controlled and reproducible setup?

Future comparison:

$$TRM_{baseline} \rightarrow TRM + HistoryAttention$$

6 Experiment Structure

Created structure:

```
experiments/  
  
  baseline  
    configs  
    results
```

Current branch:

```
phase1-baseline
```

7 Baseline Dataset

7.1 Reason for New Dataset

The previous dataset:

```
data/sudoku-tiny
```

was only suitable for smoke testing.

7.2 New Dataset

A larger baseline dataset was generated:

```
data/sudoku-small
```

Generated using the original TRM preprocessing pipeline:

```
dataset/build_sudoku_dataset.py
```

Configuration:

- Original puzzles: 1000
- Augmentation factor: 10

Approximate training examples:

$$1000 \times (1 + 10) \approx 11000$$

8 Dataset Validation

The following checks passed:

- Required files exist.
- Metadata is valid.
- Input and label shapes match.

- Token ranges are correct.
- Dataset format is compatible with TRM PuzzleDataset.

Validation result:

```
DATASET VALIDATION PASS
```

9 Baseline Configuration

Created configuration:

```
config/cfg_baseline_cpu.yaml
```

Main differences from smoke configuration:

Parameter	Smoke	Baseline
Dataset	sudoku-tiny	sudoku-small
Epochs	2	20
Batch size	1	4
Project	trm-cpu-smoke	trm-baseline
Checkpoint	smoke checkpoint	baseline-v1

Current baseline configuration:

```
data_paths:
- data/sudoku-small

global_batch_size: 4

epochs: 20

eval_interval: 5

project_name: trm-baseline

run_name: baseline-v1

checkpoint_path:
checkpoints/baseline-v1
```

10 Configuration Validation

Hydra configuration resolution was successfully tested:

```
python pretrain.py --config-name cfg_baseline_cpu --cfg job
```

Validation results:

- TRM architecture loaded successfully.
- Dataset path resolved correctly.
- Training configuration loaded correctly.

11 Dataset Loader Validation

The baseline dataset was successfully loaded:

```
all  
  
torch.Size([4,81])  
  
torch.Size([4,81])  
  
4
```

Confirmed:

- Batch generation works.
- Input shape is correct.
- Label shape is correct.

12 Current Status

Current progress:

```
Phase 0  
=====  
COMPLETE  
  
Phase 1  
=====  
IN PROGRESS
```

Completed Phase 1 tasks:

- Baseline dataset creation.
- Dataset validation.
- Baseline configuration creation.
- Hydra configuration validation.
- Dataset loader validation.

13 Next Steps

The next steps are:

1. Run a one-epoch baseline experiment.
2. Measure CPU runtime.

3. Estimate full baseline training time.
4. Run baseline-v1 experiment.
5. Store metrics and checkpoint results.

14 Development Rules

Before obtaining a stable baseline:

- Do not modify the TRM architecture.
- Do not implement HistoryAttention yet.
- Keep baseline and experimental models separated.
- Preserve reproducibility of every experiment.